Demo: Eddy Currents

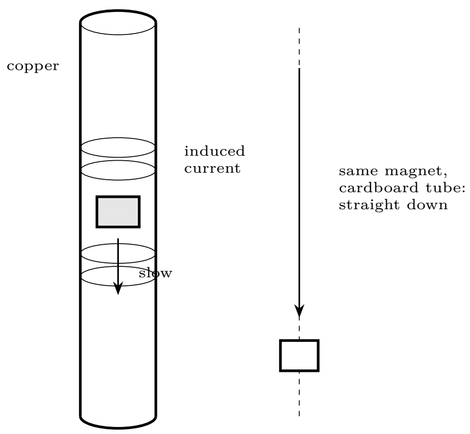
Four demonstrations of the same startling idea

Magnets falling in slow motion through metal that is not magnetic

Lesson Demo C · 20 minutes · after Lesson 5 — no power supply at all

What you need

- A copper pipe, 30–60 cm, that a magnet drops through freely
- An aluminium tube of the same size, if you can get one
- Strong neodymium magnets, and a non-magnetic slug of the same weight
- A thick aluminium or copper plate
- An aluminium pie dish and a nail, for the spinning disc
- Stopwatch



Predict first

Copper is not magnetic — a magnet will not stick to it, and you can check that first. The magnet never touches the pipe wall. So what could possibly be slowing it down? Write an answer before you drop anything, and time both falls.

What happens A falling magnet changes the magnetic field through each ring of the pipe wall as it passes. By Faraday's law that induces a voltage, and because the pipe is a good conductor, real current flows in circular loops — eddies — around the magnet.

Which way By Lenz's law, always the direction that *opposes* the change. The loop ahead of the magnet repels it; the loop behind attracts it. Both slow the fall.

Terminal velocity The braking force grows with speed, so the magnet settles at a constant slow speed rather than accelerating. Exactly like a parachute, with no air involved.

Where the energy goes Into heat in the pipe. The magnet loses gravitational energy and the copper warms up by an unmeasurably small amount. Nothing is lost.

Four versions, in this order

1. **The pipe.** Drop a magnet down copper, and a non-magnetic slug of the same weight beside it. Time both. Then aluminium, then cardboard.
2. **The pendulum.** Swing a magnet on a string just above a thick aluminium plate. It stops within a couple of swings. Lift it 2 cm and it swings freely — the effect falls off sharply with distance.
3. **The slide.** Tilt the aluminium plate and slide a strong magnet down it. It creeps. Put a sheet of paper between and it still creeps: nothing is touching.
4. **The spinning disc.** Balance an aluminium pie dish on a nail point and spin a magnet underneath it, close but not touching. The dish follows the magnet round. This is an induction motor.

Where this is used

Roller-coaster brakes	Fins of copper passing between magnets. No contact, no pads, nothing to wear out, and it cannot fail closed.
Train brakes	The same idea on a much larger scale on high-speed rail.
Your electricity meter	The old spinning-disc type is exactly demonstration 4 running backwards.
Metal detectors	They induce eddy currents in buried metal and listen for the field those currents make.
Induction hobs	Eddy currents induced directly in the pan, heating it while the hob stays cool.
Why transformers are laminated	Because here eddy currents are pure loss. Splitting the core into insulated sheets breaks the loops. *see the **Transformers** sheet*

The sentence to make them say out loud

"The magnet is being slowed down by a magnetic field that its own falling created." Nothing touches it, copper is not magnetic, and there is no battery anywhere. Once a child can say that sentence and mean it, they understand induction better than most adults.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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